

```
In [1]: import numpy as np
import pandas as pd
import seaborn as sns
```

```
In [2]: df = sns.load_dataset('titanic')
```

```
In [3]: df.head()
```

```
Out[3]:
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_m
0	0	3	male	22.0	1	0	7.2500	S	Third	man	T
1	1	1	female	38.0	1	0	71.2833	C	First	woman	Fa
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	Fa
3	1	1	female	35.0	1	0	53.1000	S	First	woman	Fa
4	0	3	male	35.0	0	0	8.0500	S	Third	man	T



```
In [5]: grouped = df.groupby('sex')['age']
# We group by sex and analyse by age
```

```
In [6]: grouped.mean()
```

```
Out[6]: sex
female    27.915709
male      30.726645
Name: age, dtype: float64
```

```
In [7]: grouped.median()
```

```
Out[7]: sex
female    27.0
male      29.0
Name: age, dtype: float64
```

```
In [9]: grouped.min()
```

```
Out[9]: sex
female    0.75
male      0.42
Name: age, dtype: float64
```

```
In [10]: grouped.max()
```

```
Out[10]: sex
female    63.0
male      80.0
Name: age, dtype: float64
```

```
In [11]: grouped.std()
```

```
# Measures spread of data.
```

```
Out[11]: sex
         female    14.110146
         male      14.678201
         Name: age, dtype: float64
```

```
In [12]: grouped.agg(['mean', 'median', 'min', 'max', 'std'])
#agg() - Aggregation function. Calculates multiple statistics together.
```

```
Out[12]:
```

	mean	median	min	max	std
sex					
female	27.915709	27.0	0.75	63.0	14.110146
male	30.726645	29.0	0.42	80.0	14.678201

```
In [16]: male_list = df[df['sex']=='male']['age'].tolist()
         female_list = df[df['sex']=='female']['age'].tolist()

         print(male_list)
         print(female_list)
```

[22.0, 35.0, nan, 54.0, 2.0, 20.0, 39.0, 2.0, nan, 35.0, 34.0, 28.0, nan, 19.0, nan, 40.0, 66.0, 28.0, 42.0, nan, 21.0, nan, nan, nan, nan, 7.0, 21.0, 65.0, nan, 28.5, 1 1.0, 22.0, 45.0, 4.0, nan, nan, 19.0, 26.0, 32.0, 21.0, 26.0, 32.0, 25.0, nan, nan, 0.83, 22.0, 29.0, 28.0, 16.0, nan, 24.0, 29.0, 20.0, 46.0, 26.0, 59.0, nan, 71.0, 2 3.0, 34.0, nan, 21.0, 33.0, 37.0, 28.0, nan, 38.0, 47.0, 22.0, 21.0, 70.5, 29.0, 24. 0, 21.0, nan, 32.5, 54.0, 12.0, nan, 24.0, 45.0, 33.0, 20.0, 25.0, 23.0, 37.0, 16.0, 24.0, 19.0, 18.0, 19.0, 27.0, 36.5, 42.0, 51.0, 55.5, 40.5, nan, 51.0, 30.0, nan, na n, 44.0, 26.0, 17.0, 1.0, 9.0, nan, 28.0, 61.0, 4.0, 21.0, 56.0, 18.0, nan, 30.0, 3 6.0, nan, 9.0, 1.0, nan, 45.0, 40.0, 36.0, 19.0, 3.0, nan, 42.0, 28.0, nan, 34.0, 4 5.5, 18.0, 32.0, 26.0, 40.0, 24.0, 22.0, 30.0, nan, 42.0, 30.0, 16.0, 27.0, 51.0, na n, 38.0, 22.0, 19.0, 20.5, 18.0, 29.0, 59.0, 24.0, 44.0, 19.0, 33.0, 29.0, 22.0, 30. 0, 44.0, 37.0, 54.0, nan, 62.0, 30.0, nan, 3.0, 52.0, 40.0, 36.0, 16.0, 25.0, nan, 2 5.0, 37.0, nan, 7.0, 65.0, 28.0, 16.0, 19.0, nan, 33.0, 30.0, 22.0, 42.0, 36.0, 24. 0, nan, 23.5, nan, nan, 19.0, nan, 0.92, 30.0, 28.0, 43.0, 54.0, 22.0, 27.0, nan, 6 1.0, 45.5, 38.0, 16.0, nan, 29.0, 45.0, 45.0, 2.0, 28.0, 25.0, 36.0, 3.0, 42.0, 23. 0, nan, 15.0, 25.0, nan, 28.0, 40.0, 29.0, 35.0, nan, 30.0, 25.0, 18.0, 19.0, 22.0, 27.0, 20.0, 19.0, 32.0, nan, 18.0, 1.0, nan, 36.0, 21.0, 28.0, 22.0, 46.0, 23.0, 39. 0, 26.0, 28.0, 34.0, 51.0, 3.0, 21.0, nan, nan, nan, 44.0, 30.0, nan, 21.0, 29.0, 1 8.0, nan, nan, 32.0, 28.0, 17.0, 50.0, 64.0, 31.0, 20.0, 25.0, nan, 4.0, 34.0, 52.0, 36.0, nan, 30.0, 49.0, nan, 29.0, 65.0, nan, 48.0, 34.0, 47.0, 48.0, nan, 38.0, nan, 56.0, nan, nan, 38.0, nan, 34.0, 29.0, 22.0, 9.0, nan, 50.0, 25.0, 58.0, 30.0, 9.0, nan, 21.0, 55.0, 71.0, 21.0, nan, nan, 24.0, 17.0, 18.0, nan, 28.0, 26.0, 29.0, nan, 36.0, 24.0, 47.0, nan, 32.0, 22.0, nan, nan, 40.5, nan, 39.0, 23.0, nan, 17.0, 45.0, nan, 32.0, 50.0, 64.0, nan, 33.0, 8.0, 17.0, 27.0, nan, 22.0, 62.0, nan, nan, 40.0, 28.0, nan, 24.0, 19.0, nan, 32.0, 62.0, 36.0, 16.0, 19.0, 32.0, 54.0, 36.0, nan, 47. 0, 60.0, 22.0, nan, 35.0, 47.0, 37.0, 36.0, 49.0, nan, 49.0, nan, nan, 44.0, 35.0, 3 6.0, 30.0, 27.0, nan, nan, 35.0, 34.0, 26.0, 27.0, 42.0, 20.0, 21.0, 21.0, 61.0, 57. 0, 26.0, nan, 80.0, 51.0, 32.0, nan, 32.0, 31.0, nan, 20.0, nan, 48.0, 19.0, 56.0, n an, nan, 21.0, 24.0, nan, 23.0, 58.0, 50.0, 40.0, 47.0, 36.0, 20.0, 32.0, 25.0, nan, 43.0, 31.0, 70.0, 31.0, nan, 18.0, 24.5, 36.0, 27.0, 20.0, 14.0, 60.0, 25.0, 14.0, 1 9.0, 18.0, 31.0, nan, 25.0, 60.0, 52.0, 44.0, 49.0, 42.0, 35.0, 25.0, 26.0, 39.0, 4 2.0, nan, nan, 48.0, 29.0, 52.0, 19.0, nan, 33.0, 17.0, 34.0, 50.0, 27.0, 20.0, 25. 0, 11.0, nan, 23.0, 23.0, 28.5, 35.0, nan, nan, nan, 36.0, 24.0, 31.0, 70.0, 16.0, 1 9.0, 31.0, 6.0, 33.0, 23.0, 0.67, 28.0, 18.0, 34.0, nan, 41.0, 20.0, 16.0, nan, nan, 32.0, 24.0, 48.0, nan, 18.0, nan, nan, 29.0, nan, 25.0, 25.0, 8.0, 1.0, 46.0, nan, 1 6.0, nan, 25.0, 39.0, 30.0, 34.0, 11.0, 0.42, 27.0, 31.0, 39.0, 39.0, 26.0, 39.0, 3 5.0, 30.5, nan, 31.0, 43.0, 10.0, 27.0, 38.0, 2.0, nan, nan, 1.0, nan, 0.83, nan, 2 3.0, 18.0, 21.0, nan, 32.0, nan, 20.0, 16.0, 34.5, 17.0, 42.0, nan, 35.0, 28.0, 4.0, 74.0, 51.0, nan, 41.0, 21.0, 24.0, 31.0, nan, 4.0, 26.0, 33.0, 47.0, 20.0, 19.0, na n, 33.0, 28.0, 25.0, 27.0, 26.0, 32.0]

[38.0, 26.0, 35.0, 27.0, 14.0, 4.0, 58.0, 14.0, 55.0, 31.0, nan, 15.0, 8.0, 38.0, na n, nan, nan, 18.0, 14.0, 40.0, 27.0, 3.0, 19.0, nan, 18.0, 49.0, 29.0, 21.0, 5.0, 3 8.0, 29.0, 17.0, 16.0, 30.0, nan, 17.0, 33.0, 23.0, 34.0, 28.0, 21.0, nan, 14.5, 20. 0, 17.0, 2.0, 32.5, nan, 47.0, 29.0, 19.0, nan, 22.0, 24.0, 9.0, 22.0, 16.0, 40.0, n an, 45.0, 1.0, 50.0, nan, 4.0, nan, 32.0, 19.0, 44.0, 58.0, nan, 24.0, 2.0, 16.0, 3 5.0, 31.0, 27.0, 32.0, nan, 35.0, 5.0, nan, 8.0, nan, nan, 25.0, 24.0, 29.0, 41.0, 2 9.0, nan, 30.0, 35.0, 50.0, nan, 58.0, 35.0, 41.0, nan, 63.0, 45.0, 35.0, 22.0, 26. 0, 19.0, 24.0, 2.0, 50.0, nan, nan, nan, 17.0, 30.0, 24.0, 18.0, 26.0, 26.0, 24.0, 3 1.0, 40.0, 30.0, 22.0, 36.0, 36.0, 31.0, 16.0, nan, nan, 41.0, 24.0, 24.0, 40.0, na n, 22.0, 38.0, nan, nan, 45.0, 60.0, nan, nan, 24.0, 3.0, nan, 22.0, 42.0, 1.0, 35. 0, 36.0, 17.0, 23.0, 24.0, 31.0, 28.0, 21.0, 20.0, nan, 33.0, nan, 34.0, 18.0, 10.0, 28.0, 28.0, 19.0, nan, 42.0, 14.0, 21.0, 24.0, 45.0, 28.0, 13.0, 5.0, nan, 50.0, 0.7 5, 33.0, 23.0, 22.0, 2.0, 63.0, nan, 35.0, 54.0, 25.0, 21.0, nan, 37.0, 16.0, 33.0, 54.0, 34.0, 36.0, 30.0, 44.0, 50.0, 2.0, nan, 30.0, 7.0, 30.0, 22.0, 36.0, 9.0, 11. 0, 19.0, 22.0, 48.0, 39.0, 36.0, nan, 29.0, 53.0, nan, 34.0, 39.0, nan, 25.0, 39.0, 18.0, 52.0, nan, nan, 24.0, 22.0, 40.0, 39.0, nan, 24.0, 26.0, 4.0, 21.0, 9.0, 28.0,

```
41.0, 24.0, 2.0, 0.75, 23.0, 18.0, nan, 18.0, 32.0, nan, 40.0, 18.0, 43.0, nan, 15.0, 4.0, nan, 18.0, 18.0, 45.0, 22.0, 24.0, 38.0, 27.0, 6.0, 30.0, nan, 25.0, 29.0, 48.0, 21.0, 30.0, 4.0, 48.0, 33.0, 36.0, 51.0, 30.5, 57.0, 54.0, 5.0, 43.0, 13.0, 17.0, 18.0, nan, 49.0, 31.0, 30.0, 31.0, 18.0, 33.0, 6.0, 23.0, 52.0, 27.0, 62.0, 15.0, 39.0, 30.0, nan, 9.0, 16.0, 44.0, 18.0, 45.0, 24.0, 48.0, nan, 42.0, 27.0, 47.0, 28.0, 15.0, 56.0, 25.0, 22.0, 39.0, 19.0, nan]
```

```
In [17]: df2 = pd.read_csv('Iris.csv')
```

```
In [18]: df2.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Id              150 non-null   int64
1   SepalLengthCm   150 non-null   float64
2   SepalWidthCm    150 non-null   float64
3   PetalLengthCm   150 non-null   float64
4   PetalWidthCm    150 non-null   float64
5   Species         150 non-null   object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

```
In [19]: df2.head()
```

```
Out[19]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
In [20]: df2['Species'].unique()
```

```
Out[20]: array(['Iris-setosa', 'Iris-versicolor', 'Iris-virginica'], dtype=object)
```

```
In [21]: setosa = df2[df2['Species']=='Iris-setosa']
```

```
In [22]: versicolor = df2[df2['Species']=='Iris-versicolor']
```

```
In [23]: virginica = df2[df2['Species']=='Iris-virginica']
```

```
In [24]: setosa.describe
```

```

Out[24]: <bound method NDFrame.describe of
Cm PetalWidthCm Species Id SepalLengthCm SepalWidthCm PetalLength
0 1 5.1 3.5 1.4 0.2 Iris-setosa
1 2 4.9 3.0 1.4 0.2 Iris-setosa
2 3 4.7 3.2 1.3 0.2 Iris-setosa
3 4 4.6 3.1 1.5 0.2 Iris-setosa
4 5 5.0 3.6 1.4 0.2 Iris-setosa
5 6 5.4 3.9 1.7 0.4 Iris-setosa
6 7 4.6 3.4 1.4 0.3 Iris-setosa
7 8 5.0 3.4 1.5 0.2 Iris-setosa
8 9 4.4 2.9 1.4 0.2 Iris-setosa
9 10 4.9 3.1 1.5 0.1 Iris-setosa
10 11 5.4 3.7 1.5 0.2 Iris-setosa
11 12 4.8 3.4 1.6 0.2 Iris-setosa
12 13 4.8 3.0 1.4 0.1 Iris-setosa
13 14 4.3 3.0 1.1 0.1 Iris-setosa
14 15 5.8 4.0 1.2 0.2 Iris-setosa
15 16 5.7 4.4 1.5 0.4 Iris-setosa
16 17 5.4 3.9 1.3 0.4 Iris-setosa
17 18 5.1 3.5 1.4 0.3 Iris-setosa
18 19 5.7 3.8 1.7 0.3 Iris-setosa
19 20 5.1 3.8 1.5 0.3 Iris-setosa
20 21 5.4 3.4 1.7 0.2 Iris-setosa
21 22 5.1 3.7 1.5 0.4 Iris-setosa
22 23 4.6 3.6 1.0 0.2 Iris-setosa
23 24 5.1 3.3 1.7 0.5 Iris-setosa
24 25 4.8 3.4 1.9 0.2 Iris-setosa
25 26 5.0 3.0 1.6 0.2 Iris-setosa
26 27 5.0 3.4 1.6 0.4 Iris-setosa
27 28 5.2 3.5 1.5 0.2 Iris-setosa
28 29 5.2 3.4 1.4 0.2 Iris-setosa
29 30 4.7 3.2 1.6 0.2 Iris-setosa
30 31 4.8 3.1 1.6 0.2 Iris-setosa
31 32 5.4 3.4 1.5 0.4 Iris-setosa
32 33 5.2 4.1 1.5 0.1 Iris-setosa
33 34 5.5 4.2 1.4 0.2 Iris-setosa
34 35 4.9 3.1 1.5 0.1 Iris-setosa
35 36 5.0 3.2 1.2 0.2 Iris-setosa
36 37 5.5 3.5 1.3 0.2 Iris-setosa
37 38 4.9 3.1 1.5 0.1 Iris-setosa
38 39 4.4 3.0 1.3 0.2 Iris-setosa
39 40 5.1 3.4 1.5 0.2 Iris-setosa
40 41 5.0 3.5 1.3 0.3 Iris-setosa
41 42 4.5 2.3 1.3 0.3 Iris-setosa
42 43 4.4 3.2 1.3 0.2 Iris-setosa
43 44 5.0 3.5 1.6 0.6 Iris-setosa
44 45 5.1 3.8 1.9 0.4 Iris-setosa
45 46 4.8 3.0 1.4 0.3 Iris-setosa
46 47 5.1 3.8 1.6 0.2 Iris-setosa
47 48 4.6 3.2 1.4 0.2 Iris-setosa
48 49 5.3 3.7 1.5 0.2 Iris-setosa
49 50 5.0 3.3 1.4 0.2 Iris-setosa>

```

```

In [26]: setosa.shape

```

Out[26]: (50, 6)

In [27]: `versicolor.describe`

```

Out[27]: <bound method NDFrame.describe of
hCm PetalWidthCm \ Id SepalLengthCm SepalWidthCm PetalLengt
50 51 7.0 3.2 4.7 1.4
51 52 6.4 3.2 4.5 1.5
52 53 6.9 3.1 4.9 1.5
53 54 5.5 2.3 4.0 1.3
54 55 6.5 2.8 4.6 1.5
55 56 5.7 2.8 4.5 1.3
56 57 6.3 3.3 4.7 1.6
57 58 4.9 2.4 3.3 1.0
58 59 6.6 2.9 4.6 1.3
59 60 5.2 2.7 3.9 1.4
60 61 5.0 2.0 3.5 1.0
61 62 5.9 3.0 4.2 1.5
62 63 6.0 2.2 4.0 1.0
63 64 6.1 2.9 4.7 1.4
64 65 5.6 2.9 3.6 1.3
65 66 6.7 3.1 4.4 1.4
66 67 5.6 3.0 4.5 1.5
67 68 5.8 2.7 4.1 1.0
68 69 6.2 2.2 4.5 1.5
69 70 5.6 2.5 3.9 1.1
70 71 5.9 3.2 4.8 1.8
71 72 6.1 2.8 4.0 1.3
72 73 6.3 2.5 4.9 1.5
73 74 6.1 2.8 4.7 1.2
74 75 6.4 2.9 4.3 1.3
75 76 6.6 3.0 4.4 1.4
76 77 6.8 2.8 4.8 1.4
77 78 6.7 3.0 5.0 1.7
78 79 6.0 2.9 4.5 1.5
79 80 5.7 2.6 3.5 1.0
80 81 5.5 2.4 3.8 1.1
81 82 5.5 2.4 3.7 1.0
82 83 5.8 2.7 3.9 1.2
83 84 6.0 2.7 5.1 1.6
84 85 5.4 3.0 4.5 1.5
85 86 6.0 3.4 4.5 1.6
86 87 6.7 3.1 4.7 1.5
87 88 6.3 2.3 4.4 1.3
88 89 5.6 3.0 4.1 1.3
89 90 5.5 2.5 4.0 1.3
90 91 5.5 2.6 4.4 1.2
91 92 6.1 3.0 4.6 1.4
92 93 5.8 2.6 4.0 1.2
93 94 5.0 2.3 3.3 1.0
94 95 5.6 2.7 4.2 1.3
95 96 5.7 3.0 4.2 1.2
96 97 5.7 2.9 4.2 1.3
97 98 6.2 2.9 4.3 1.3
98 99 5.1 2.5 3.0 1.1
99 100 5.7 2.8 4.1 1.3

```

Species

```

50 Iris-versicolor
51 Iris-versicolor

```

```
52 Iris-versicolor
53 Iris-versicolor
54 Iris-versicolor
55 Iris-versicolor
56 Iris-versicolor
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60 Iris-versicolor
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88 Iris-versicolor
89 Iris-versicolor
90 Iris-versicolor
91 Iris-versicolor
92 Iris-versicolor
93 Iris-versicolor
94 Iris-versicolor
95 Iris-versicolor
96 Iris-versicolor
97 Iris-versicolor
98 Iris-versicolor
99 Iris-versicolor >
```

```
In [28]: versicolor.shape
```

```
Out[28]: (50, 6)
```

```
In [29]: virginica.describe
```



```

Out[29]: <bound method NDFrame.describe of
thCm  PetalWidthCm  \
100  101          6.3          3.3          6.0          2.5
101  102          5.8          2.7          5.1          1.9
102  103          7.1          3.0          5.9          2.1
103  104          6.3          2.9          5.6          1.8
104  105          6.5          3.0          5.8          2.2
105  106          7.6          3.0          6.6          2.1
106  107          4.9          2.5          4.5          1.7
107  108          7.3          2.9          6.3          1.8
108  109          6.7          2.5          5.8          1.8
109  110          7.2          3.6          6.1          2.5
110  111          6.5          3.2          5.1          2.0
111  112          6.4          2.7          5.3          1.9
112  113          6.8          3.0          5.5          2.1
113  114          5.7          2.5          5.0          2.0
114  115          5.8          2.8          5.1          2.4
115  116          6.4          3.2          5.3          2.3
116  117          6.5          3.0          5.5          1.8
117  118          7.7          3.8          6.7          2.2
118  119          7.7          2.6          6.9          2.3
119  120          6.0          2.2          5.0          1.5
120  121          6.9          3.2          5.7          2.3
121  122          5.6          2.8          4.9          2.0
122  123          7.7          2.8          6.7          2.0
123  124          6.3          2.7          4.9          1.8
124  125          6.7          3.3          5.7          2.1
125  126          7.2          3.2          6.0          1.8
126  127          6.2          2.8          4.8          1.8
127  128          6.1          3.0          4.9          1.8
128  129          6.4          2.8          5.6          2.1
129  130          7.2          3.0          5.8          1.6
130  131          7.4          2.8          6.1          1.9
131  132          7.9          3.8          6.4          2.0
132  133          6.4          2.8          5.6          2.2
133  134          6.3          2.8          5.1          1.5
134  135          6.1          2.6          5.6          1.4
135  136          7.7          3.0          6.1          2.3
136  137          6.3          3.4          5.6          2.4
137  138          6.4          3.1          5.5          1.8
138  139          6.0          3.0          4.8          1.8
139  140          6.9          3.1          5.4          2.1
140  141          6.7          3.1          5.6          2.4
141  142          6.9          3.1          5.1          2.3
142  143          5.8          2.7          5.1          1.9
143  144          6.8          3.2          5.9          2.3
144  145          6.7          3.3          5.7          2.5
145  146          6.7          3.0          5.2          2.3
146  147          6.3          2.5          5.0          1.9
147  148          6.5          3.0          5.2          2.0
148  149          6.2          3.4          5.4          2.3
149  150          5.9          3.0          5.1          1.8

```

```

Species
100  Iris-virginica
101  Iris-virginica

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```
In [30]: virginica.shape
```

```
Out[30]: (50, 6)
```

Other way :

```
iris.groupby('Species').describe()
```

Measures of Central Tendency Central tendency represents center of data. Types:

Mean Average value. Sensitive to outliers.

Median Middle value. Robust against outliers.

Mode Most frequent value. Used for categorical data.

Measures of Dispersion Show variability/spread of data. Types:

Range

Variance

Standard deviation

Variance Measures average squared deviation from mean. Formula: $\sigma^2 = \frac{\sum (x - \mu)^2}{N}$

Standard Deviation Square root of variance. Formula: $\sigma = \sqrt{\sigma^2}$

Why Standard Deviation Important? Shows consistency of data. Low SD:

Values close to mean

High SD:

Values widely spread

Important Viva Questions

Basic Questions What is mean? Average of observations.

Difference between mean and median? MeanMedianAverageMiddle valueAffected by outliersLess affected

What is variance? Average squared deviation from mean.

What is standard deviation? Square root of variance.

What does percentile mean? Position of value in distribution.

Intermediate Questions Why median preferred sometimes? Less affected by outliers.

What is grouping in statistics? Dividing data into categories.

What is aggregation? Combining multiple statistical operations.

Why use describe()? Quick statistical summary.

External May Ask Modifications

Calculate Mode `df['Age'].mode()`

Calculate Variance `df['Age'].var()`

Group By Another Variable `df.groupby('Pclass')['Fare'].mean()`

Find Percentiles `df['Age'].quantile([0.25,0.5,0.75])`

Find Correlation `df.corr(numeric_only=True)`

Common Viva Tricky Questions Which measure is most affected by outliers? Mean.

Which measure is least affected by outliers? Median.

Why variance squared? To avoid cancellation of positive and negative deviations.

Why square root in standard deviation? To bring units back to original scale.

2-Minute Viva Explanation "In this practical, we calculated basic statistical measures like mean, median, minimum, maximum, variance, standard deviation, and percentiles using Python Pandas. First, we grouped Titanic dataset records based on categorical variable 'Sex' and calculated summary statistics of Age. Then using the Iris dataset, we filtered flower species and generated statistical summaries using `describe()`. Measures of central tendency help identify central values, while measures of dispersion help understand variability in the data."